Hyperspectral Image Denoising and Destriping via Gradient Tensor Subspace Low-Rank Learning and Along-Across Stripe Directional Constraints

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***Abstract*— This article proposes a new hyperspectral image (HSI) denoising and destriping method via gradient tensor subspace low-rank learning and along-across stripe directional constraints (GTSL2A2SDCs) under the unified framework of tensor representation modeling. On one hand, for the modeling of stripe noise, based on the inherently directional and structural attributes of the stripe noise, we mainly investigate the mode-1 gradient tensor of stripe along the stripe direction which holds the preferable "zero plane" constraint as well as the mode-2 gradient tensor of stripe across the stripe direction which holds the preferable tensor low-fibered-rank attribute along the mode-3 spectral dimension. Therefore, we novelly propose the unified along-across stripe directional gradient tensor constraints for the stripe noise, which can simultaneously characterize the directional and structural attributes of the stripe noise. On the other hand, for the modeling of HSI, based on the preferably spectral low-rankness attributes of the multimode gradient tensors of HSI, namely, mode-1 gradient tensor, mode-2 gradient tensor, and mode-3 gradient tensor, we further utilize the spectral low-rank factorization of the gradient tensors of HSI to get the corresponding representation tensors, and particularly investigate the nonlocal self-similarities (NSSs)-based low-rankness of the representation tensors under the gradient tensor-based subspace low-rank learning framework. Moreover, we optimize the proposed GTSL2A2SDC model via an efficiently alternative and iterative algorithm. At last, extensive experiments comprehensively validate the denoising and destriping capacity and superiority of GTSL2A2SDC.**

***Index Terms*— Along-across stripe directional gradient tensor constraints, hyperspectral image (HSI) denoising and destriping, multimode gradient tensor subspace low-rank learning, representation tensors, spectral low-rankness, tensor low-fibered-rank.**

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NOMENCLATURE

a, $\mathbf{a}$, $\mathbf{A}$, $\mathcal{A}$	Scalar, vector, matrix, tensor.
$\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$	HQ HSI (a third-order tensor).
$\mathcal{Y} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$	LQ noisy HSI (a third-order tensor).
$\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$	Stripe noise (a third-order tensor).
$\mathcal{S} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$	Sparse noise (a third-order tensor) (i.e., impulse noise and deadlines).
$\mathcal{X}(i_1, i_2, i_3)$	(i_1, i_2, i_3)th element of $\mathcal{X}$.
$\mathcal{X}(:, :, i_3)$	i_3th frontal slice of $\mathcal{X}$.
$\text{unfold}_3(\mathcal{X}) \in \mathbb{R}^{n_3 \times n_1 n_2}$	Mode-3 unfolding of $\mathcal{X}$.
$\mathbf{X}_{(3)} \in \mathbb{R}^{n_3 \times n_1 n_2}$	Mode-3 unfolding of $\mathcal{X}$.
$\lvert \cdot \rvert$	Absolute value operator.
$\text{rank}(\cdot)$	Rank operator of matrix.
$\lVert \cdot \rVert_*$	Nuclear norm operator of matrix.
$\lVert \mathcal{X} \rVert_1 = \sum_{i_1,i_2,i_3} \lvert \mathcal{X}(i_1, i_2, i_3) \rvert$	l_1 norm of $\mathcal{X}$.
$\lVert \mathcal{X} \rVert_F = (\sum_{i_1,i_2,i_3} \lvert \mathcal{X}(i_1, i_2, i_3) \rvert^2)^{1/2}$	Frobenius norm of $\mathcal{X}$.
$(\cdot)^T$	Transpose operator.

I. INTRODUCTION

IN THE field of remote sensing, the hyperspectral images (HSIs) with rich and discriminative spectral information are extensively used for agroforestry, target detection, military reconnaissance, geologic survey, and disaster monitoring [1]. Nevertheless, the HSIs during imaging acquisition are usually and unavoidably polluted by diversely mixed noise, such as, Gaussian noise, stripe noise, impulse noise, and deadlines, which significantly leads to the reduction of quality of HSIs as well as the limitation of applications of HSIs. To acquire high-quality (HQ) HSIs for meeting the demands of applications yet becomes a more urgent task recently. Therefore, the effective HSI mixed denoising technique is developed to remove the mixed noise from the low-quality (LQ) noisy HSIs so as to obtain the HQ HSIs, which has been an important and challenging topic in remote sensing.

A. Related Works: HSI Mixed Denoising

Up to now, researchers had proposed many different types of HSI mixed denoising methods, such as, model-driven methods, data-driven methods, and model-data-driven methods [2].

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Specifically, the data-driven methods and the model-data-driven methods [2] belong to the deep learning-based methods for HSI mixed denoising. Moreover, there are various and representative methods among them, such as, the convolutional neural network (CNN)-based methods [3], [4], [5], [6], the transformer-based methods [7], [8], and the model-guided deep networks-based methods [9], [10], [11], [12]. Generally, they can show advanced denoising performance, however, their generalization abilities are not stable for the HSIs with different and complex mixed noise. Please see [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12] for more details about the data-driven methods and the model-data-driven methods, which are not the scope of this article. More specifically, this article mainly aims to study the model-driven HSI mixed denoising methods.

Apparently, the task of HSI mixed denoising is to recover the latent HQ HSI from the LQ noisy HSI, which is seriously contaminated by the Gaussian noise, impulse noise, deadlines, and stripe noise. Mathematically, based on the tensor representation modeling, the LQ noisy HSI is modeled as follows:

$$\mathcal{Y} = \mathcal{X} + \mathcal{S} + \mathcal{N} \tag{1}$$

where $\mathcal{Y} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ denotes the LQ noisy HSI, $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ denotes the latent HQ HSI to be recovered, $\mathcal{S} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ denotes the mixed impulse noise, deadlines, and stripe noise, $\mathcal{N} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ denotes the Gaussian noise.

Thus, the model-driven HSI mixed denoising methods aim to exploit the latent prior knowledge of HSI and diverse mixed noise to mathematically construct a unified minimization optimization model as follows:

$$\left(\hat{\mathcal{X}}, \hat{\mathcal{S}}\right) = \arg\min_{\mathcal{X},\mathcal{S}} \frac{1}{2} \| \mathcal{Y} - \mathcal{X} - \mathcal{S} \|_F^2 + \beta R(\mathcal{S}) + \alpha J(\mathcal{X}) \tag{2}$$

where $(1/2)\|\mathcal{Y} - \mathcal{X} - \mathcal{S}\|_F^2$ aims to model the Gaussian noise $\mathcal{N}$, $R(\mathcal{S})$ denotes the prior of mixed impulse noise, deadlines, and stripe noise $\mathcal{S}$, $J(\mathcal{X})$ denotes the prior of HSI $\mathcal{X}$, β, and α denote the regularization parameters.

More apparently, the model-driven HSI mixed denoising methods aim to exploit the diverse priors of HSI and mixed noise, which usually either treat the stripe noise together with impulse noise and deadlines as also the sparse noise for sparse prior modeling, or treat the stripe noise separately different from the impulse noise and deadlines for stripe prior modeling.

On one hand, by treating the stripe noise together with impulse noise and deadlines as wholly the sparse noise $\mathcal{S} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ for sparse prior modeling, the priors of HSI can be mainly classified as follows.

1) TV-based priors: The total variation (TV)-based priors aim to explore the spatial and spectral gradients-based local smoothness attributes of HSI for denoising. Among them, cubic TV [13], spectral–spatial adaptive hyperspectral TV [14], multidimensional nonlocal TV [15], enhanced 3-D TV [16], and representative coefficient TV (RCTV) [17] are very well-known. Clearly, the TV-based methods can well preserve image edges, however, they usually fail to preserve image details owing to TV favors the piecewise constant solution.

2) Low-Rank Priors: The low-rank priors aim to explore the latent spectral and spatial low-rankness of HSI, which are usually modeled as the low-rank matrix priors or the low-rank tensor priors.

For the low-rank matrix priors-based methods, they usually explore the spectral low-rankness of HSI by unfolding the HSI along mode-3 to be the low-rank matrix, and explore the nonlocal low-rankness of HSI by modeling each similar nonlocal patch group to be the low-rank matrix. Among them, there are many representative methods, such as, low-rank matrix recovery (LRMR) [18], noise-adjusted iterative low-rank matrix approximation [19], global and nonlocal low-rank factorizations [20], nonlocal meets global (NGmeet) [21], factor group sparsity-regularized nonconvex low-rank approximation [22], and correntropy-based nonconvex low-rank approximation [23]. Clearly, the low-rank matrix priors-based methods can well preserve image details and structures, however, they heavily destroy the spectral and spatial structures of the 3-D HSI $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, which is actually the third-order tensor.

Thus, for preserving the spectral and spatial structures of the 3-D HSI $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, the low-rank tensor priors-based methods are proposed to explore the tensor-based low-rank priors by modeling the HSI as the third-order tensor and using low-rank tensor decomposition and tensor rank approximation techniques. Among them, there are also many representative methods, such as, nonlocal low-rank tensor decomposition [24], nonlocal tensor-ring decomposition [25], group sparsity-regularized low-rank tensor decomposition (LRTDGS) [26], low-tensor-ring-rank prior in the transformed domain [27], 3-D tensor nuclear norm (3DTNN) [28], weighted-order-p tensor nuclear norm (TNN) [29], spatial gradient tensor low-rank prior [30], low-rank Tucker decomposition [31], and guidable nonlocal low-rank approximation model [32]. Specifically, the low-rank tensor priors-based methods can better preserve the 3-D structures of HSI.

3) Hybrid TV and Low-Rank Priors: Besides, the hybrid TV and low-rank priors-based denoising methods are also proposed to explore the TV priors and low-rank priors of HSI together for better preserving structures and details of HSI, which are either modeled as the combined TV and low-rank regularization terms [33], [34], [35], [36], [37], [38], [39], [40] or modeled as the unified joint local smoothness and low-rankness regularization terms [41], [42]. Specifically, these methods can perform promising denoising results.

B. Related Works: HSI Denoising and Destriping

Indeed, different from the sparse impulse noise and deadlines, the stripe noise $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ has inherent directionality and structural property [43], [44], [45], [46], [47], [48]. Therefore, the structural priors of stripe noise need to be separately explored unlike the sparse noise.

On the other hand, by treating the stripe noise separately different from the impulse noise and deadlines for stripe prior modeling owing to the strong structural property of stripe noise, many HSI denoising and destriping methods are recently proposed, which aim to explore the structural priors

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of HSI and the stripe noise at the same time. More clearly, for the prior modeling of stripe noise, the stripe noise is usually and simply modeled by the band-wise low-rank priors [46], [49], [50], [51], [52], such as, $\sum_{i_3=1}^{n_3} \text{rank}(\mathcal{B}(:,:,i_3))$ and $\sum_{i_3=1}^{n_3} \|\mathcal{B}(:,:,i_3)\|_*$. Moreover, based on the third-order tensor formulation of stripe noise, only several recent methods explore the tensor-based low-rank priors of stripe noise [47], [48]. However, these methods only explore the low-rank priors of stripe noise in the original domain, and ignore to fully explore the strong directionality of stripe noise, where the stripe noise actually has the strong directionality.

C. Our Contributions

Therefore, in this article, to overcome these shortcomings, we propose to fully explore the strong directionality and structural property of stripe noise in the gradient domain via gradient tensor-based directional constraints. More specifically, from the points of view of multimode gradient tensor-based subspace low-rank learning of HSI as well as directional and structural prior modeling of stripe noise, this article proposes a new HSI denoising and destriping method via gradient tensor subspace low-rank learning and along-across stripe directional constraints (GTSL2A2SDCs) under the unified framework of tensor representation modeling. Apparently, the major contributions of this article are listed as below.

1) **On one hand**, for the modeling of the stripe noise, based on the inherently directional and structural attributes of the stripe noise, we mainly investigate the mode-1 gradient tensor of stripe along the stripe direction which holds the preferable "zero-plane" constraint as well as the mode-2 gradient tensor of stripe across the stripe direction which holds the preferable tensor low-fibered-rank attribute along the mode-3 spectral dimension. Therefore, we novelly propose the unified along-across stripe directional gradient tensor constraints for the stripe noise, which can simultaneously characterize the directional and structural attributes of the stripe noise.
2) **On the other hand**, for the modeling of HSI, based on the preferably spectral low-rankness attributes of the multimode gradient tensors of HSI, namely, mode-1 gradient tensor, mode-2 gradient tensor, and mode-3 gradient tensor, we further utilize the spectral low-rank factorization of the multimode gradient tensors of HSI to get the corresponding representation tensors, and particularly investigate the nonlocal self-similarities (NSSs)-based low-rankness of the representation tensors under the multimode gradient tensor subspace low-rank learning framework.
3) Furthermore, by integrating the multimode gradient tensor subspace low-rank learning-based nonlocal prior of HSI and the unified along-across stripe directional gradient tensor constraints of stripe noise, we construct a new GTSL2A2SDC model for HSI denoising and destriping. Then, we optimize the proposed GTSL2A2SDC model via the alternating direction method of multipliers (ADMMs). At last, extensive experiments comprehensively validate the denoising and destriping capacity and superiority of GTSL2A2SDC.

The rest sections of this article are listed as follows. In Section II, we construct the GTSL2A2SDC model. In Section III, we optimize the proposed GTSL2A2SDC model. In Section IV, we show the experimental results. In Section V, we show the conclusions.

II. Proposed GTSL2A2SDC Model

A. Symbols and Definitions

First, we give lots of important symbols and definitions in Nomenclature, which will be used throughout this article for modeling and analysis.

Definition 1 (Gradient Tensor [42]): Given a third-order tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, its gradient tensor along ith mode, i.e., $\nabla_i \mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, $i = 1, 2, 3$, is defined as follows:

$$\nabla_i \mathcal{A} = \mathcal{A} \times_i \mathbf{D}_{n_i}, \quad i = 1, 2, 3 \tag{3}$$

where ∇_i denotes the gradient operator along the ith mode, $\mathcal{A} \times_i \mathbf{D}_{n_i}$ denotes the mode-i product of tensor $\mathcal{A}$ with matrix $\mathbf{D}_{n_i}$, and $\mathbf{D}_{n_i} \in \mathbb{R}^{n_i \times n_i}$ denotes the row circulant matrix of $(-1, 1, 0, \ldots, 0) \in \mathbb{R}^{1 \times n_i}$, which is formulated as follows:

$$\mathbf{D}_{n_i} = \begin{bmatrix} -1 & 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & -1 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \ddots & \ddots & \\ 0 & 0 & 0 & 0 & \cdots & -1 & 1 \\ 1 & 0 & 0 & 0 & \cdots & 0 & -1 \end{bmatrix}_{n_i \times n_i}. \tag{4}$$

B. Formulation of Model-Driven HSI Denoising and Destriping

Apparently, the task of simultaneously HSI denoising and destriping is to recover the latent HQ HSI from the LQ noisy HSI, which is seriously contaminated by the Gaussian noise, sparse noise (i.e., impulse noise and deadlines), and stripe noise. Mathematically, based on the tensor representation modeling, the LQ noisy HSI is modeled as follows:

$$\mathcal{Y} = \mathcal{X} + \mathcal{S} + \mathcal{B} + \mathcal{N} \tag{5}$$

where $\mathcal{Y} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ denotes the LQ noisy HSI, $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ denotes the latent HQ HSI to be recovered, $\mathcal{S} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, and $\mathcal{N} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ separately denote the sparse noise, stripe noise, and Gaussian noise.

More apparently, for the problem of simultaneously HSI denoising and destriping, we independently and contrapuntally model the stripe noise in this article. Thus, the model-driven simultaneously HSI denoising and destriping model is generally formulated as follows:

$$\begin{aligned} (\hat{\mathcal{X}}, \hat{\mathcal{S}}, \hat{\mathcal{B}}) = \arg\min_{\mathcal{X},\mathcal{S},\mathcal{B}} & \frac{1}{2}\|\mathcal{Y} - \mathcal{X} - \mathcal{S} - \mathcal{B}\|_F^2 + \beta\|\mathcal{S}\|_1 \\ & + \lambda R(\mathcal{B}) + \alpha J(\mathcal{X}) \end{aligned} \tag{6}$$

where $(1/2)\|\mathcal{Y} - \mathcal{X} - \mathcal{S} - \mathcal{B}\|_F^2$ aims to model the Gaussian noise $\mathcal{N}$, $\|\mathcal{S}\|_1$ denotes the sparse prior of sparse noise $\mathcal{S}$, $R(\mathcal{B})$ denotes the prior of stripe noise $\mathcal{B}$, $J(\mathcal{X})$ denotes the prior of HSI $\mathcal{X}$, and β, λ, and α denote the regularization parameters.

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C. Along-Across Stripe Directional Gradient Tensor Constraints

In this section, by viewing the stripe noise $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ separately as a third-order tensor in this article, we aim to simultaneously investigate the directional and structural prior of the stripe noise $\mathcal{B}$ in a unified tensor modeling framework.

Apparently, the stripe noise $\mathcal{B}$ not only has the inherently directional attribute, but also has the inherently structural attribute. However, the inherently directional and structural attributes of the stripe noise are not fully and simultaneously investigated in the previous methods, in which they model the stripe noise $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ either as together the sparse noise $\mathcal{S} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ using the sparse prior term $\|\mathcal{S}\|_1$ or as separately the stripe noise using the band-wise low-rank prior terms [46], [49], [50], [51], [52], such as, $\sum_{i_3=1}^{n_3} \text{rank}(\mathcal{B}(:,:,i_3))$ and $\sum_{i_3=1}^{n_3} \|\mathcal{B}(:,:,i_3)\|_*$. Moreover, based on the third-order tensor formulation of stripe noise, only several recent methods explore the tensor-based low-rank priors of stripe noise [47], [48]. However, these methods only explore the low-rank priors of stripe noise in the original domain, and ignore to fully explore the strong directionality of stripe noise, where the stripe noise actually has the strong directionality.

Thus, we aim to simultaneously investigate the directional and structural prior of the stripe noise $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ in a unified tensor modeling framework, and novelly propose the unified along-across stripe directional gradient tensor constraints for the stripe noise $\mathcal{B}$.

On one hand, based on the directional attribute of stripe noise $\mathcal{B}$, which is often vertical, we propose to investigate the mode-1 gradient tensor of stripe noise $\mathcal{B}$ along the stripe based on Definition 1, i.e., $\nabla_1\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, and clearly have the "zero plane" constraint of $\nabla_1\mathcal{B}$ along the stripe (see Fig. 1), which is formulated as follows:

$$\nabla_1\mathcal{B} = \mathcal{O} \tag{7}$$

where $\mathcal{O} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ denotes the tensor with all elements of zero.

Remark 1: The mode-1 gradient tensor of stripe noise $\mathcal{B}$ satisfies $\nabla_1\mathcal{B} = \mathcal{O}$, and we call it as "zero plane" constraint of $\nabla_1\mathcal{B}$, mainly owing to each slice of $\nabla_1\mathcal{B}$ [i.e., $\nabla_1\mathcal{B}(:,:,i_3)$] is actually a zero matrix, where each pixel of $\nabla_1\mathcal{B}(:,:,i_3)$ locates in the zero plane.

Furthermore, we novelly define an indicator function $\Phi(\nabla_1\mathcal{B})$ for above "zero plane" constraint (7) of $\nabla_1\mathcal{B}$ along the stripe such that

$$\Phi(\nabla_1\mathcal{B}) = \begin{cases} 0, & \text{if } \nabla_1\mathcal{B} = \mathcal{O} \\ +\infty, & \text{otherwise.} \end{cases} \tag{8}$$

On the other hand, based on the structural attribute of stripe noise $\mathcal{B}$, which is often low-rank, we propose to investigate the mode-2 gradient tensor of stripe noise $\mathcal{B}$ across the stripe based on Definition 1, i.e., $\nabla_2\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$. Furthermore, we employ the mode-3 tensor singular value decomposition (t-SVD) [28] of $\nabla_2\mathcal{B}$, and clearly have the tensor mode-3 low-fibered-rank attribute of $\nabla_2\mathcal{B}$ across the stripe (see Fig. 1), i.e., the mode-3 tensor fibered rank of $\nabla_2\mathcal{B}$ denoted as $\text{rank}_{f_3}(\nabla_2\mathcal{B})$ is low, which is defined as [28]

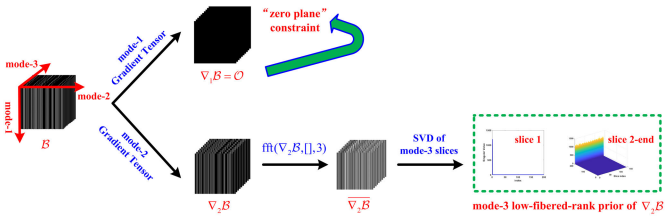


Fig. 1. Flowchart of the unified along-across stripe directional gradient tensor constraints for the stripe noise $\mathcal{B}$.

$$\text{rank}_{f_3}(\nabla_2\mathcal{B}) = \sharp\{i_3, \mathcal{D}_3(:,:,i_3) \neq \mathbf{O}\} \tag{9}$$

i.e., the number of nonzero mode-3 fibers of $\mathcal{D}_3$, where $\nabla_2\mathcal{B} = \mathcal{C}_3 *_3 \mathcal{D}_3 *_3 \mathcal{E}_3^T$ denotes the mode-3 t-SVD of $\nabla_2\mathcal{B}$, $\sharp\{\cdot\}$ denotes the cardinality operator of a set, and $\mathbf{O} \in \mathbb{R}^{n_1 \times n_2}$ denotes the matrix with all elements of zero.

Furthermore, the tensor mode-3 low-fibered-rank prior term of $\nabla_2\mathcal{B}$ is proposed as follows:

$$\mathcal{R}_1(\mathcal{B}) = \text{rank}_{f_3}(\nabla_2\mathcal{B}). \tag{10}$$

However, the direct minimization of the $\text{rank}_{f_3}(\nabla_2\mathcal{B})$ is NP-hard. Thus, we utilize the mode-3 TNN [28] for the tensor low-rank approximation of $\text{rank}_{f_3}(\nabla_2\mathcal{B})$, and propose the mode-3 TNN-based tensor low-fibered-rank prior term of $\nabla_2\mathcal{B}$ as follows:

$$\mathcal{R}_1(\mathcal{B}) = \|\nabla_2\mathcal{B}\|_{\text{TNN}_3} \tag{11}$$

where the mode-3 TNN of $\nabla_2\mathcal{B}$ is defined as $\|\nabla_2\mathcal{B}\|_{\text{TNN}_3} = \sum_{i_3=1}^{n_3} \|\overline{\nabla_2\mathcal{B}}(:,:,i_3)\|_*$ and $\overline{\nabla_2\mathcal{B}} = \text{fft}(\nabla_2\mathcal{B}, [\,], 3)$.

Thus, the unified along-across stripe directional gradient tensor constraints for the stripe noise $\mathcal{B}$ are constructed as follows:

$$\mathcal{R}(\mathcal{B}) = \lambda\|\nabla_2\mathcal{B}\|_{\text{TNN}_3} + \Phi(\nabla_1\mathcal{B}). \tag{12}$$

D. Multimode Gradient Tensor-Based Subspace Low-Rank Learning Prior of HSI

As we clearly know, the HSI $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ in its original domain has desirable low-rankness attribute along the spectral dimension, which is also called spectral low-rankness of $\mathcal{X}$. Thus, the mode-3 unfolding matrix of $\mathcal{X}$, i.e., $\text{unfold}_3(\mathcal{X}) \in \mathbb{R}^{n_3 \times n_1 n_2}$ (or $\mathbf{X}_{(3)} \in \mathbb{R}^{n_3 \times n_1 n_2}$), is low-rank, where $\text{rank}(\mathbf{X}_{(3)}) = r \ll n_3$ is the rank of $\mathbf{X}_{(3)}$.

More different from the original domain of HSI $\mathcal{X}$, in this article, we focus on investigating the multimode gradient domain of HSI $\mathcal{X}$. Based on Definition 1, we can obtain the gradient tensor of HSI $\mathcal{X}$ along the ith mode, i.e., $\nabla_i\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, $i = 1, 2, 3$.

Moreover, we know that the gradient operator ∇_i along the ith mode is linear and has the full-rank attribute, thus the gradient tensors $\nabla_1\mathcal{X}$, $\nabla_2\mathcal{X}$, and $\nabla_3\mathcal{X}$ hold the same spectral low-rankness attributes as $\mathcal{X}$ along the spectral dimension, i.e., $\text{unfold}_3(\nabla_i\mathcal{X}) \in \mathbb{R}^{n_3 \times n_1 n_2}$ (or $\nabla_i\mathbf{X}_{(3)} \in \mathbb{R}^{n_3 \times n_1 n_2}$), $i = 1, 2, 3$, are also low-rank, which can be also validated by Fig. 2. Meanwhile, $\text{rank}(\nabla_1\mathbf{X}_{(3)}) = \text{rank}(\nabla_2\mathbf{X}_{(3)}) = \text{rank}(\nabla_3\mathbf{X}_{(3)}) = \text{rank}(\mathbf{X}_{(3)}) = r \ll n_3$ clearly holds.

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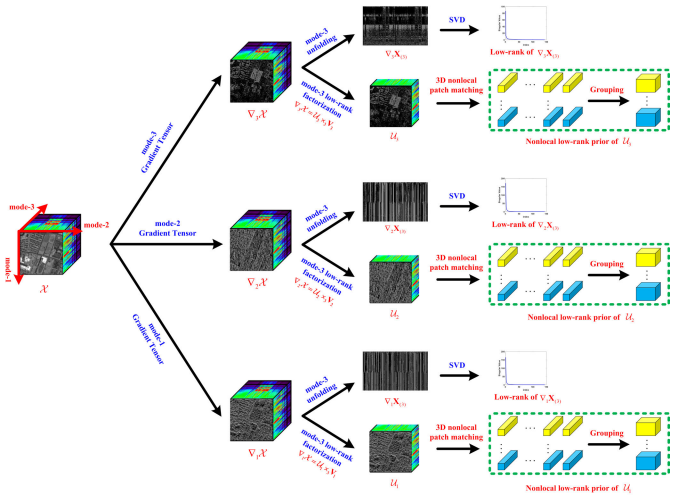


Fig. 2. Flowchart of the unified multimode gradient tensor subspace low-rank learning-based nonlocal prior of HSI $\mathcal{X}$.

Depended on the spectral low-rankness attributes of the gradient tensors $\nabla_1\mathcal{X}$, $\nabla_2\mathcal{X}$, and $\nabla_3\mathcal{X}$ along the spectral dimension, we decide to employ the spectral low-rank factorization of $\nabla_1\mathcal{X}$, $\nabla_2\mathcal{X}$, and $\nabla_3\mathcal{X}$ into the latent subspace, and have the corresponding gradient tensor subspace representation as follows:

$$\nabla_i\mathcal{X} = \mathcal{U}_i \times_3 \mathbf{V}_i, \quad i = 1, 2, 3 \tag{13}$$

where $\mathcal{U}_i \in \mathbb{R}^{n_1\times n_2\times r}$ is the representation tensor, and $\mathbf{V}_i \in \mathbb{R}^{n_3\times r}$ is the spectral orthogonal basis satisfying $\mathbf{V}_i^T\mathbf{V}_i = \mathbf{I} \in \mathbb{R}^{r\times r}$ and $\text{rank}(\mathbf{V}_i) = r \ll n_3$.

Thus, depended on the preferable spectral low-rank factorization of gradient tensors $\nabla_i\mathcal{X}$ in (13), we aim to deeply study the latent structural attributes of the representation tensor $\mathcal{U}_i$, $i = 1, 2, 3$, in the gradient tensor-based subspace.

Specifically, the representation tensor $\mathcal{U}_i \in \mathbb{R}^{n_1\times n_2\times r}$ apparently not only has rich and important structural information as the HSI gradient tensor $\nabla_i\mathcal{X}$, but also has strong NSS (see Fig. 2). Thus, we propose to explore the 3-D patches-based NSS of representation tensor $\mathcal{U}_i$ and impose the nonlocal low-rank prior of $\mathcal{U}_i$ as follows:

$$J(\mathcal{U}_i) = \|\mathcal{U}_i\|_{\text{NL}} = \sum_j \psi\left(\mathcal{R}_j\mathcal{U}_i\right) \tag{14}$$

where $\mathcal{R}_j\mathcal{U}_i$ denotes the jth exemplar 3-D patch group consisting of its similar patches extracted from $\mathcal{U}_i$, and $\psi(\cdot)$ denotes the low-rank approximation function.

Thus, based on the gradient tensor-based spectral low-rank factorization $\nabla_i\mathcal{X} = \mathcal{U}_i \times_3 \mathbf{V}_i$, $\mathbf{V}_i^T\mathbf{V}_i = \mathbf{I}$, $i = 1, 2, 3$ in (13) and the nonlocal low-rank prior of representation tensor $\mathcal{U}_i$ in (14), we construct the unified gradient tensor subspace low-rank learning-based nonlocal prior of HSI $\mathcal{X}$ as follows:

$$\begin{aligned} J(\mathcal{X}) = \min_{\mathcal{U}_i} \frac{\eta}{2}\sum_{i=1}^{3}\|\nabla_i\mathcal{X} - \mathcal{U}_i \times_3 \mathbf{V}_i\|_F^2 + \alpha\sum_{i=1}^{3}\|\mathcal{U}_i\|_{\text{NL}} \\ \text{s.t. } \mathbf{V}_i^T\mathbf{V}_i = \mathbf{I}, \quad i = 1, 2, 3 \end{aligned} \tag{15}$$

where η is the trade-off parameter.

E. Proposed GTSL2A2SDC Model

Subsequently, by integrating the unified along-across stripe directional gradient tensor constraints for the stripe noise $\mathcal{B}$ in (12) and the unified gradient tensor subspace low-rank learning-based nonlocal prior of HSI $\mathcal{X}$ in (15) into model (6), we propose a new HSI denoising and destriping model via GTSL2A2SDCs as follows:

$$\begin{aligned} &(\hat{\mathcal{X}}, \hat{\mathcal{S}}, \hat{\mathcal{B}}, \hat{\mathcal{U}}_i, \hat{\mathbf{V}}_i) \\ &= \arg\min_{\mathcal{X},\mathcal{S},\mathcal{B},\mathcal{U}_i,\mathbf{V}_i} \frac{1}{2}\|\mathcal{Y} - \mathcal{X} - \mathcal{S} - \mathcal{B}\|_F^2 + \beta\|\mathcal{S}\|_1 \\ &\quad + \frac{\eta}{2}\sum_{i=1}^{3}\|\nabla_i\mathcal{X} - \mathcal{U}_i \times_3 \mathbf{V}_i\|_F^2 + \alpha\sum_{i=1}^{3}\|\mathcal{U}_i\|_{\text{NL}} \\ &\quad + \lambda\|\nabla_2\mathcal{B}\|_{\text{TNN}_3} + \Phi(\nabla_1\mathcal{B}) \\ &\quad \text{s.t.} \quad \mathbf{V}_i^T\mathbf{V}_i = \mathbf{I}, \quad i = 1, 2, 3. \end{aligned} \tag{16}$$

III. Algorithm

Depended on the ADMM framework, we optimize the proposed GTSL2A2SDC model (16) by first introducing two auxiliary tensor variables $\mathcal{G} \in \mathbb{R}^{n_1\times n_2\times n_3}$ and $\mathcal{P} \in \mathbb{R}^{n_1\times n_2\times n_3}$ such that $\mathcal{G} = \nabla_2\mathcal{B}$ and $\mathcal{P} = \nabla_1\mathcal{B}$, and then convert the model (16) as follows:

$$\begin{aligned} \min_{\mathcal{X},\mathcal{S},\mathcal{B},\mathcal{G},\mathcal{P},\mathcal{U}_i,\mathbf{V}_i} &\frac{1}{2}\|\mathcal{Y}-\mathcal{X}-\mathcal{S}-\mathcal{B}\|_F^2 + \beta\|\mathcal{S}\|_1 \\ &+\frac{\eta}{2}\sum_{i=1}^{3}\|\nabla_i\mathcal{X}-\mathcal{U}_i \times_3 \mathbf{V}_i\|_F^2 + \alpha\sum_{i=1}^{3}\|\mathcal{U}_i\|_{\text{NL}} \\ &+\lambda\|\mathcal{G}\|_{\text{TNN}_3} + \Phi(\mathcal{P}) \\ \text{s.t.} \quad &\mathbf{V}_i^T\mathbf{V}_i = \mathbf{I}, \quad i = 1, 2, 3 \\ &\mathcal{G} = \nabla_2\mathcal{B} \\ &\mathcal{P} = \nabla_1\mathcal{B}. \end{aligned} \tag{17}$$

Furthermore, the augmented Lagrange function of (17) is expressed as follows:

$$\begin{aligned} &f(\mathcal{X}, \mathcal{S}, \mathcal{B}, \mathcal{G}, \mathcal{P}, \mathcal{U}_i, \mathbf{V}_i, \mathcal{H}, \mathcal{Q}) \\ &= \frac{1}{2}\|\mathcal{Y} - \mathcal{X} - \mathcal{S} - \mathcal{B}\|_F^2 + \beta\|\mathcal{S}\|_1 \\ &\quad + \sum_{i=1}^{3}\left(\frac{\eta}{2}\|\nabla_i\mathcal{X} - \mathcal{U}_i \times_3 \mathbf{V}_i\|_F^2 + \alpha\|\mathcal{U}_i\|_{\text{NL}}\right) \\ &\quad + \frac{\mu}{2}\|\mathcal{G} - \nabla_2\mathcal{B} - \mathcal{H}\|_F^2 + \lambda\|\mathcal{G}\|_{\text{TNN}_3} \\ &\quad + \frac{\rho}{2}\|\mathcal{P} - \nabla_1\mathcal{B} - \mathcal{Q}\|_F^2 + \Phi(\mathcal{P}) \end{aligned} \tag{18}$$

in which $\mathcal{H} \in \mathbb{R}^{n_1\times n_2\times n_3}$ and $\mathcal{Q} \in \mathbb{R}^{n_1\times n_2\times n_3}$ are Lagrangian multipliers, and μ and ρ are penalty parameters.

By setting $\mathcal{X}^{(0)}$, $\mathcal{S}^{(0)}$, $\mathcal{B}^{(0)}$, $\mathcal{G}^{(0)}$, $\mathcal{P}^{(0)}$, $\mathcal{U}_i^{(0)}$, $\mathbf{V}_i^{(0)}$, $\mathcal{H}^{(0)}$, and $\mathcal{Q}^{(0)}$ $(i = 1, 2, 3)$, the model (17) is optimized by the following subproblems at $(k+1)$th iteration.

A. Optimization of $\mathcal{S}$

Based on (18), the $\mathcal{S}^{(k+1)}$ subproblem is derived as follows:

$$\mathcal{S}^{(k+1)} = \arg\min_{\mathcal{S}} \frac{1}{2}\left\|\mathcal{Y} - \mathcal{X}^{(k)} - \mathcal{S} - \mathcal{B}^{(k)}\right\|_F^2 + \beta\|\mathcal{S}\|_1 \tag{19}$$

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which is easily solved by the soft-thresholding (ST) approach as follows:

$$\mathcal{S}^{(k+1)} = \mathfrak{S}_{\beta}\left(\mathcal{Y} - \mathcal{X}^{(k)} - \mathcal{B}^{(k)}\right) \tag{20}$$

where the ST operator is $\mathfrak{S}_{\beta}(\mathcal{Z}) = \text{sgn}(\mathcal{Z})\max(|\mathcal{Z}| - \beta)$.

B. Optimization of $\mathcal{B}$

Based on (18), the $\mathcal{B}^{(k+1)}$ subproblem is derived as follows:

$$\begin{aligned}\mathcal{B}^{(k+1)} = \arg\min_{\mathcal{B}} \frac{1}{2}\left\|\mathcal{Y} - \mathcal{X}^{(k)} - \mathcal{S}^{(k+1)} - \mathcal{B}\right\|_F^2 \\ + \frac{\mu}{2}\left\|\mathcal{G}^{(k)} - \nabla_2\mathcal{B} - \mathcal{H}^{(k)}\right\|_F^2 \\ + \frac{\rho}{2}\left\|\mathcal{P}^{(k)} - \nabla_1\mathcal{B} - \mathcal{Q}^{(k)}\right\|_F^2\end{aligned} \tag{21}$$

which is equivalent to solving the following linear equation:

$$\begin{aligned}\left(\mathcal{I} + \rho\,\nabla_1^T\nabla_1 + \mu\,\nabla_2^T\nabla_2\right)\mathcal{B} = \mathcal{Y} - \mathcal{X}^{(k)} - \mathcal{S}^{(k+1)} \\ + \rho\nabla_1^T\left(\mathcal{P}^{(k)} - \mathcal{Q}^{(k)}\right) \\ + \mu\nabla_2^T\left(\mathcal{G}^{(k)} - \mathcal{H}^{(k)}\right).\end{aligned} \tag{22}$$

Therefore, we solve (22) by applying the fast Fourier transform (FT) scheme and get $\mathcal{B}^{(k+1)}$ as follows:

$$\mathcal{B}^{(k+1)} = \text{ifftn}\left(\frac{\text{fftn}(\mathcal{J} + \rho\,\mathcal{K} + \mu\,\mathcal{L})}{1 + \rho|\text{fftn}(\nabla_1)|^2 + \mu|\text{fftn}(\nabla_2)|^2}\right) \tag{23}$$

where $\mathcal{J} = \mathcal{Y} - \mathcal{X}^{(k)} - \mathcal{S}^{(k+1)}$, $\mathcal{K} = \nabla_1^T(\mathcal{P}^{(k)} - \mathcal{Q}^{(k)})$, $\mathcal{L} = \nabla_2^T(\mathcal{G}^{(k)} - \mathcal{H}^{(k)})$, $\text{fftn}(\cdot)$ denotes the 3-D FT operator, and $\text{ifftn}(\cdot)$ denotes the inverse 3-D FT operator.

C. Optimization of $\mathcal{G}$

Based on (18), the $\mathcal{G}^{(k+1)}$ subproblem is derived as follows:

$$\mathcal{G}^{(k+1)} = \arg\min_{\mathcal{G}} \frac{\mu}{2}\left\|\mathcal{G} - \nabla_2\mathcal{B}^{(k+1)} - \mathcal{H}^{(k)}\right\|_F^2 + \lambda\|\mathcal{G}\|_{\text{TNN}_3} \tag{24}$$

which is optimized by the mode-3 TNN-based tensor singular value thresholding (t-SVT) approach [28] as follows:

$$\mathcal{G}^{(k+1)} = \mathfrak{D}_{\text{TNN}_3}^{\lambda/\mu}\left(\nabla_2\mathcal{B}^{(k+1)} + \mathcal{H}^{(k)}\right) \tag{25}$$

where the mode-3 TNN-based t-SVT operator $\mathfrak{D}_{\text{TNN}_3}^{\tau}(\cdot)$ is defined as follows:

$$\mathfrak{D}_{\text{TNN}_3}^{\tau}(\mathcal{Z}) = \mathcal{C} *_3 \mathcal{D}^{\tau} *_3 \mathcal{E}^T \tag{26}$$

where $\mathcal{Z} = \mathcal{C} *_3 \mathcal{D} *_3 \mathcal{E}^T$ is the mode-3 t-SVD of $\mathcal{Z}$, $\bar{\mathcal{D}}^{\tau} = \max(\bar{\mathcal{D}} - \tau, 0)$, $\bar{\mathcal{D}} = \text{fft}(\mathcal{D}, [\,], 3)$, $\mathcal{D}^{\tau} = \text{ifft}(\bar{\mathcal{D}}^{\tau}, [\,], 3)$.

D. Optimization of $\mathcal{P}$

Based on (18), the $\mathcal{P}^{(k+1)}$ subproblem is derived as follows:

$$\mathcal{P}^{(k+1)} = \arg\min_{\mathcal{P}} \frac{\rho}{2}\left\|\mathcal{P} - \nabla_1\mathcal{B}^{(k+1)} - \mathcal{Q}^{(k)}\right\|_F^2 + \Phi(\mathcal{P}). \tag{27}$$

Based on the indicator function $\Phi(\cdot)$ defined in (8), we clearly obtain $\mathcal{P}^{(k+1)}$ as follows:

$$\mathcal{P}^{(k+1)} = \mathcal{O}. \tag{28}$$

E. Optimization of $\mathcal{U}_i$

Based on (18), the $\mathcal{U}_i^{(k+1)}$ subproblem ($i = 1, 2, 3$) is derived as follows:

$$\mathcal{U}_i^{(k+1)} = \arg\min_{\mathcal{U}_i} \frac{\eta}{2}\left\|\nabla_i\mathcal{X}^{(k)} - \mathcal{U}_i \times_3 \mathbf{V}_i^{(k)}\right\|_F^2 + \alpha\,\|\mathcal{U}_i\|_{\text{NL}}. \tag{29}$$

Based on $\mathbf{V}_i^{(k)^T}\mathbf{V}_i^{(k)} = \mathbf{I}$, we can further have

$$\mathcal{U}_i^{(k+1)} = \arg\min_{\mathcal{U}_i} \frac{\eta}{2}\left\|\mathcal{U}_i - \nabla_i\mathcal{X}^{(k)} \times_3 \mathbf{V}_i^{(k)^T}\right\|_F^2 + \alpha\,\|\mathcal{U}_i\|_{\text{NL}}. \tag{30}$$

More specifically, the $\mathcal{U}_i^{(k+1)}$ subproblem (30) becomes the nonlocal low-rank prior-based denoising problem. Thus, in this article, we apply the famous and efficient weighted nuclear norm minimization (WNNM) method [53] to solve it as follows:

$$\mathcal{U}_i^{(k+1)} = \text{WNNM}\left(\nabla_i\mathcal{X}^{(k)} \times_3 \mathbf{V}_i^{(k)^T}, \alpha/\eta\right). \tag{31}$$

F. Optimization of $\mathbf{V}_i$

Based on (18), we can formulate the $\mathbf{V}_i^{(k+1)}$ subproblem ($i = 1, 2, 3$) as follows:

$$\mathbf{V}_i^{(k+1)} = \arg\min_{\mathbf{V}_i, \mathbf{V}_i^T\mathbf{V}_i = \mathbf{I}} \frac{\eta}{2}\left\|\nabla_i\mathcal{X}^{(k)} - \mathcal{U}_i^{(k+1)} \times_3 \mathbf{V}_i\right\|_F^2. \tag{32}$$

Equivalently, we further optimize the $\mathbf{V}_i^{(k+1)}$ subproblem (32) as follows:

$$\mathbf{V}_i^{(k+1)} = \arg\max_{\mathbf{V}_i, \mathbf{V}_i^T\mathbf{V}_i = \mathbf{I}} \text{Tr}\left(\mathbf{W}_i^{(k)}\mathbf{V}_i\right) \tag{33}$$

where $\mathbf{W}_i^{(k)} = \text{unfold}_3(\mathcal{U}_i^{(k+1)})(\text{unfold}_3(\nabla_i\mathcal{X}^{(k)}))^T \in \mathbb{R}^{r\times n_3}$, $i = 1, 2, 3$, and $\text{Tr}(\cdot)$ denotes the trace operator of the matrix.

Specifically, the $\mathbf{V}_i^{(k+1)}$ subproblem (33) becomes the classic orthogonal Procrustes problem [54], which has the closed-form solution as follows:

$$\mathbf{V}_i^{(k+1)} = \mathbf{E}_i^{(k)}\,\mathbf{C}_i^{(k)^T} \tag{34}$$

where $\mathbf{W}_i^{(k)} = \mathbf{C}_i^{(k)}\mathbf{D}_i^{(k)}\mathbf{E}_i^{(k)^T}$ is the singular value decomposition of $\mathbf{W}_i^{(k)}$.

G. Optimization of $\mathcal{X}$

Based on (18), the $\mathcal{X}^{(k+1)}$ subproblem is derived as follows:

$$\begin{aligned}\mathcal{X}^{(k+1)} = \arg\min_{\mathcal{X}} \frac{1}{2}\left\|\mathcal{Y} - \mathcal{X} - \mathcal{S}^{(k+1)} - \mathcal{B}^{(k+1)}\right\|_F^2 \\ + \frac{\eta}{2}\sum_{i=1}^{3}\left\|\nabla_i\mathcal{X} - \mathcal{U}_i^{(k+1)} \times_3 \mathbf{V}_i^{(k+1)}\right\|_F^2\end{aligned} \tag{35}$$

which is equivalent to solving the following linear equation:

$$\begin{aligned}\left(\mathcal{I} + \eta\sum_{i=1}^{3}\nabla_i^T\nabla_i\right)\mathcal{X} = \mathcal{Y} - \mathcal{S}^{(k+1)} - \mathcal{B}^{(k+1)} \\ + \eta\sum_{i=1}^{3}\nabla_i^T\left(\mathcal{U}_i^{(k+1)} \times_3 \mathbf{V}_i^{(k+1)}\right).\end{aligned} \tag{36}$$

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Algorithm 1 GTSL2A2SDC

Input: Noisy HSI $\mathcal{Y} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, parameters β, α, η, λ, μ, ρ, *MaxIter*.

Initialization: $k = 0$, $\mathcal{X}^{(0)} = \mathcal{Y}$, $\mathcal{S}^{(0)} = \mathcal{B}^{(0)} = \mathcal{G}^{(0)} = \mathcal{P}^{(0)} = \mathcal{U}_i^{(0)} = \mathcal{H}^{(0)} = \mathcal{Q}^{(0)} = \mathcal{O}$, $\mathbf{V}_i^{(0)} = \mathbf{O}$, $i = 1, 2, 3$, where $\mathcal{O} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is the tensor of all elements of zero, $\mathbf{O} \in \mathbb{R}^{n_3 \times r}$ is the matrix of all elements of zero.

Iterations:

For $k = 0$**to***MaxIter* $- 1$ **Do**
  Optimize $\mathcal{S}^{(k+1)}$ via (20)
  Optimize $\mathcal{B}^{(k+1)}$ via (23)
  Optimize $\mathcal{G}^{(k+1)}$ via (25)
  Optimize $\mathcal{P}^{(k+1)}$ via (28)
  Optimize $\mathcal{U}_i^{(k+1)}$ via (31)
  Optimize $\mathbf{V}_i^{(k+1)}$ via (32)-(34)
  Optimize $\mathcal{X}^{(k+1)}$ via (37)
  Update $\mathcal{H}^{(k+1)}$ and $\mathcal{Q}^{(k+1)}$ via (38)
End For

Output: Denoised HSI $\mathcal{X}^{(MaxIter)}$.

Therefore, similar to the $\mathcal{B}^{(k+1)}$ subproblem (23), we solve (36) by also applying the fast FT scheme and get $\mathcal{X}^{(k+1)}$ as follows:

$$\mathcal{X}^{(k+1)} = \text{ifftn}\left(\frac{\text{fftn}\left(\mathcal{Y} - \mathcal{S}^{(k+1)} - \mathcal{B}^{(k+1)} + \eta\,\mathcal{M}\right)}{1 + \eta \sum_{i=1}^{3} |\text{fftn}(\nabla_i)|^2}\right) \tag{37}$$

where $\mathcal{M} = \sum_{i=1}^{3} \nabla_i^T (\mathcal{U}_i^{(k+1)} \times_3 \mathbf{V}_i^{(k+1)})$.

H. Update $\mathcal{H}$ and $\mathcal{Q}$

The Lagrange multipliers $\mathcal{H}^{(k+1)}$ and $\mathcal{Q}^{(k+1)}$ are updated by the following equation:

$$\begin{cases} \mathcal{H}^{(k+1)} = \mathcal{H}^{(k)} + \nabla_2 \mathcal{B}^{(k+1)} - \mathcal{G}^{(k+1)} \\ \mathcal{Q}^{(k+1)} = \mathcal{Q}^{(k)} + \nabla_1 \mathcal{B}^{(k+1)} - \mathcal{P}^{(k+1)}. \end{cases} \tag{38}$$

At last, we also summarize the proposed GTSL2A2SDC algorithm in Algorithm 1.

IV. Experiments

A. Experimental Setting

To validate the denoising performance of the proposed GTSL2A2SDC method, we conduct the denoising experiments by using the simulated and real HSI datasets, respectively. Specifically, the GTSL2A2SDC method is compared with various HSI denoising methods, such as, LRMR [18], TV regularized low-rank tensor decomposition (LRTDTV) [37], LRTDGS [26], 3DTNN [28], weighted nonlocal low-rank model and adaptive TV (WNLRATV) [35], RCTV [17], 3-D-correlated TV (3DCTV) [41], fast graph Laplacian regularization (FGLR) [50], NGmeet [21], and fast and parameter-free HSI mixed (FastHyMix) noise removal method [10].

In detail, the simulated denoising experiments are conducted by using the Houston and Pavia city center (PaviaC) datasets, where the simulated noisy HSI is generated by adding various cases of Gaussian noise, impulse noise, deadlines, and stripe noise to the clean HSI. Moreover, the spectral angle mapper (SAM), mean peak signal to noise ratio (MPSNR), and mean structural similarity (MSSIM) are applied for quantitatively assessing the simulated denoising experiments.

What is more, the real denoising experiments are conducted by directly using the real noisy Indian pines and GaoFen-5 Shanghai datasets. More specifically, in the experiments, the intensities of HSI are normalized to [0, 1], and all the HSI denoising approaches are tested and implemented by using MATLAB(R2018b) on a personal computer equipped with Windows 10 and 2.20 GHz Intel[1] Core[2] i7-8750H CPU and 16 GB RAM.

B. Simulated Denoising Experiments

1) Simulated Noise Cases: In the simulated denoising experiments, we use the Houston dataset (size: 200 × 200 × 144) and the PaviaC dataset (size: 200 × 200 × 80). Specifically, we add four representative cases of mixed noise to the clean HSI to generate the noisy HSI for the simulated denoising experiments on the Houston and PaviaC datasets, i.e.,

Case 1 (Gaussian noise + stripe noise): First, we randomly add the Gaussian noise of standard deviation $\sigma \in [0.1, 0.2]$ to each band of HSI. Then, we further randomly add the stripe noise to 40% of bands of HSI and 30% of columns of these bands.

Case 2 (Gaussian noise + stripe noise + impulse noise): On the basis of the noise Case 1, the impulse noise of density of 1 is further randomly added to 20% bands of HSI.

Case 3 (Gaussian noise + stripe noise+ deadlines): On the basis of the noise Case 1, the deadlines are further randomly added to 11–30 bands with the stripe noise, where the number and the width of deadlines are set from the ranges of [1, 5] and [1, 3], respectively.

Case 4 (Gaussian noise + stripe noise + impulse noise + deadlines): First, the same Gaussian noise, stripe noise, and deadlines are added on the basis of noise Case 3. Then, the same impulse noise are further added on the basis of noise Case 2.

2) Quantitative and Visual Results: On one hand, Table I gives the quantitative SAM, MPSNR, and MSSIM results of different approaches on the simulated Houston and PaviaC datasets with the best bolded and the second-best underlined. Apparently, the proposed GTSL2A2SDC approach gives the best SAM and MSSIM results for all the noise cases, the best MPSNR result for the noise Case 3 of PaviaC dataset, the third-best MPSNR result for the noise Case 4 of Houston dataset, and the second-best MPSNR results for all the other noise cases so as to quantitatively show very promising performance to remove the mixed noise.

On the other hand, for visually validating the mixed noise removal performance of different approaches, in Figs. 3–8, we, respectively, show the detailed experimental results from the aspects of denoising results, pseudo-color results, and spectrum curves before and after denoising on the simulated

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TABLE I
QUANTITATIVE RESULTS OF SAM, MPSNR, AND MSSIM ON THE SIMULATED HOUSTON AND PAVIAC DATASETS

Dataset	Case	Index	Noisy	LRMR	LRTDTV	LRTDGS	3DTNN	WNLRATV	RCTV	NGmeet	3DCTV	FGLR	FastHyMix	GTSL2A2SDC
Houston	Case 1	SAM	29.9110	11.4939	10.4361	13.3746	13.7186	9.7557	10.5517	9.3001	10.7026	10.0448	9.8538	**8.3851**
		MPSNR	15.84	26.98	25.97	26.58	26.08	28.29	28.07	28.95	27.44	26.88	**29.26**	28.96
		MSSIM	0.2785	0.8321	0.7664	0.7743	0.7435	0.8352	0.8729	0.8848	0.8426	0.8367	0.8703	**0.9026**
	Case 2	SAM	29.1251	10.2696	9.0893	12.2782	12.2733	8.7282	9.1291	8.5277	8.9129	9.3561	9.0162	**7.5058**
		MPSNR	15.97	27.34	26.14	26.88	26.42	27.77	28.37	**29.18**	27.77	27.36	28.99	29.08
		MSSIM	0.2839	0.8415	0.7659	0.7840	0.7655	0.8300	0.8787	0.8869	0.8497	0.8400	0.8754	**0.9028**
	Case 3	SAM	30.0589	11.6965	10.0114	13.5432	14.2084	9.6265	10.5052	9.7143	10.2286	10.1724	10.0407	**8.5248**
		MPSNR	15.80	26.92	25.82	26.60	26.02	27.83	28.04	28.79	27.39	26.86	**29.17**	28.84
		MSSIM	0.2785	0.8293	0.7650	0.7776	0.7404	0.8240	0.8710	0.8773	0.8427	0.8369	0.8636	**0.9004**
	Case 4	SAM	29.3438	10.8064	8.8080	12.5850	13.2556	8.9991	9.5396	8.8864	9.0105	9.5884	9.4140	**7.8447**
		MPSNR	15.95	27.03	26.18	26.68	26.09	27.51	28.15	28.93	27.71	27.10	**29.00**	28.83
		MSSIM	0.2843	0.8340	0.7749	0.7803	0.7462	0.8174	0.8775	0.8840	0.8545	0.8412	0.8667	**0.9015**
PaviaC	Case 1	SAM	33.9819	9.4982	8.8592	11.1027	10.0039	7.9867	8.6803	8.5998	8.9280	6.6617	8.4920	**5.0777**
		MPSNR	16.41	27.94	27.75	28.46	27.86	28.91	29.24	30.58	28.41	29.33	**31.12**	30.94
		MSSIM	0.2483	0.8243	0.7891	0.8311	0.7927	0.8445	0.8657	0.8929	0.8380	0.8677	0.8933	**0.9096**
	Case 2	SAM	33.1840	9.4217	8.1435	11.1100	9.4654	8.6042	8.6279	9.2287	8.6883	6.5245	8.2659	**4.9282**
		MPSNR	16.71	28.10	28.19	28.46	28.06	29.26	29.19	30.17	28.64	29.48	**31.25**	30.97
		MSSIM	0.2591	0.8258	0.7998	0.8297	0.8071	0.8492	0.8631	0.8763	0.8408	0.8692	0.8998	**0.9083**
	Case 3	SAM	34.0081	9.7319	8.6012	13.2071	11.3569	7.6525	8.6116	8.4094	8.6687	6.4191	8.9166	**4.8014**
		MPSNR	16.40	27.77	27.74	28.10	27.38	29.64	29.05	30.55	28.36	29.18	30.65	**30.90**
		MSSIM	0.2492	0.8156	0.7884	0.7789	0.7642	0.8539	0.8593	0.8826	0.8365	0.8636	0.8738	**0.9076**
	Case 4	SAM	33.4615	9.4376	8.2918	12.2843	10.0989	7.8974	8.3036	9.0441	8.4615	6.3724	8.5018	**4.7647**
		MPSNR	16.66	28.06	28.21	28.17	27.88	29.23	29.17	30.16	28.67	29.51	**31.05**	31.04
		MSSIM	0.2573	0.8246	0.7991	0.7860	0.7962	0.8504	0.8670	0.8742	0.8404	0.8688	0.8892	**0.9081**

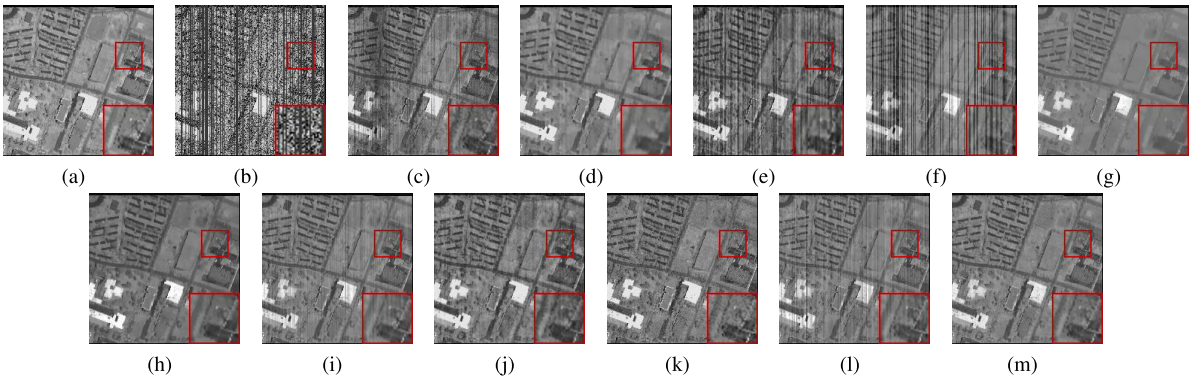


Fig. 3. Simulated denoising results of different approaches on the Houston dataset under Case 4 (band: 82). (a) Original image. (b) Noisy image. (c) LRMR. (d) LRTDTV. (e) LRTDGS. (f) 3DTNN. (g) WNLRATV. (h) RCTV. (i) NGmeet. (j) 3DCTV. (k) FGLR. (l) FastHyMix. (m) GTSL2A2SDC.

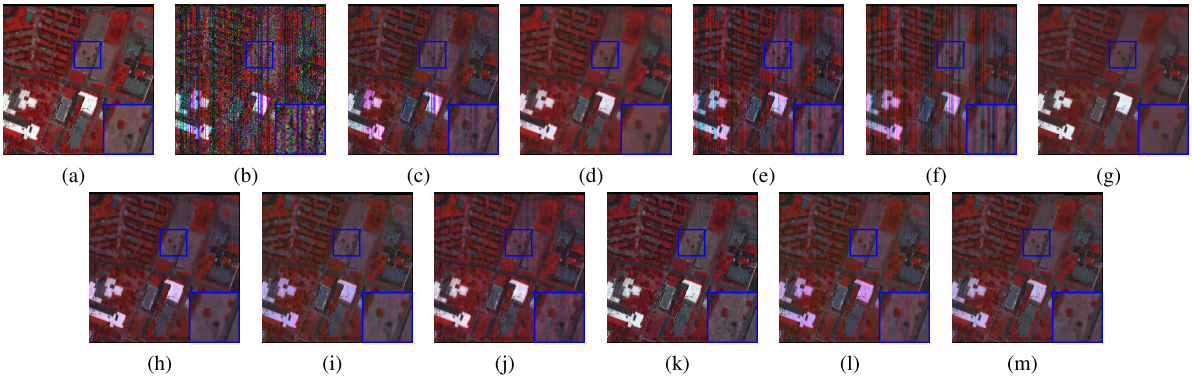


Fig. 4. Pseudo-color results of different denoising approaches on the simulated Houston dataset under Case 4 (bands: [116, 34, 33]). (a) Original image. (b) Noisy image. (c) LRMR. (d) LRTDTV. (e) LRTDGS. (f) 3DTNN. (g) WNLRATV. (h) RCTV. (i) NGmeet. (j) 3DCTV. (k) FGLR. (l) FastHyMix. (m) GTSL2A2SDC.

Houston and PaviaC datasets particularly under Case 4, which is the most serious case of mixed noise.

With the denoised images and zoomed details shown in Figs. 3, 4, 6, and 7, LRMR and 3DCTV could not absolutely

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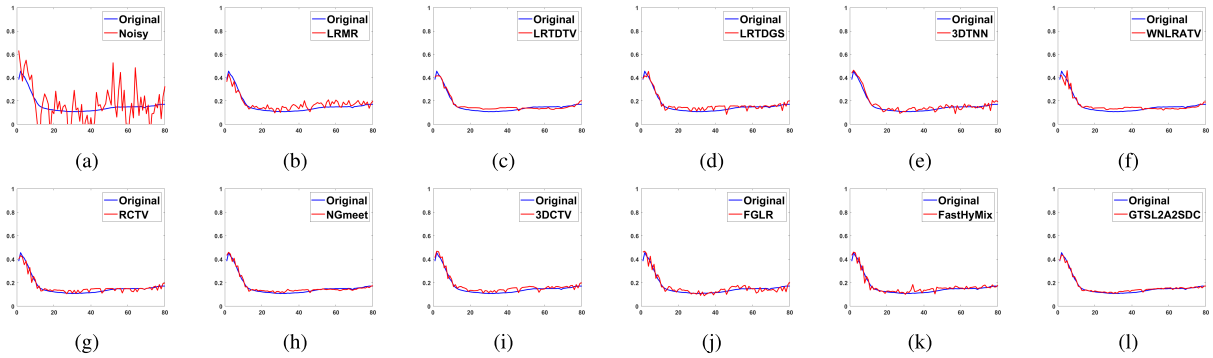


Fig. 5. Comparisons of the spectrum curves before and after denoising on the simulated Houston dataset under Case 4 at spatial position (12, 77). (a) Noisy image. (b) LRMR. (c) LRTDTV. (d) LRTDGS. (e) 3DTNN. (f) WNLRATV. (g) RCTV. (h) NGmeet. (i) 3DCTV. (j) FGLR. (k) FastHyMix. (l) GTSL2A2SDC.

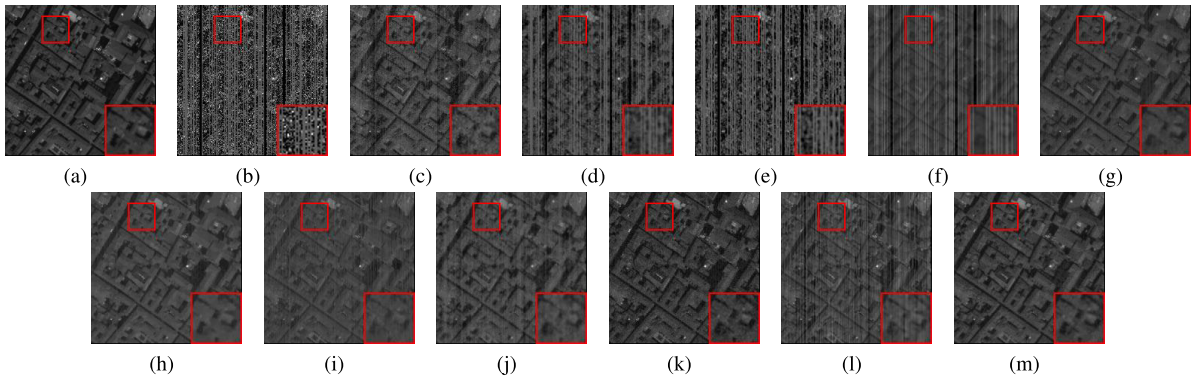


Fig. 6. Simulated denoising results of different approaches on the PaviaC dataset under Case 4 (band: 17). (a) Original image. (b) Noisy image. (c) LRMR. (d) LRTDTV. (e) LRTDGS. (f) 3DTNN. (g) WNLRATV. (h) RCTV. (i) NGmeet. (j) 3DCTV. (k) FGLR. (l) FastHyMix. (m) GTSL2A2SDC.

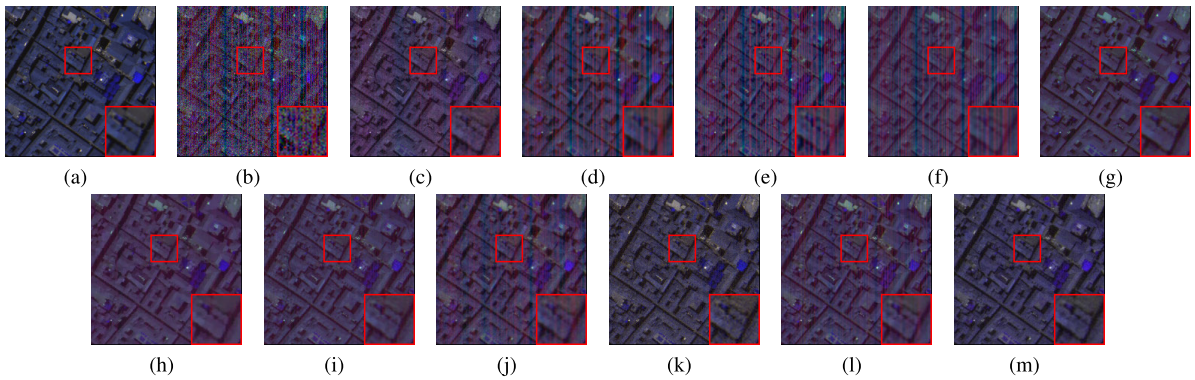


Fig. 7. Pseudo-color results of different denoising approaches on the simulated PaviaC dataset under Case 4 (bands: [6, 14, 70]). (a) Original image. (b) Noisy image. (c) LRMR. (d) LRTDTV. (e) LRTDGS. (f) 3DTNN. (g) WNLRATV. (h) RCTV. (i) NGmeet. (j) 3DCTV. (k) FGLR. (l) FastHyMix. (m) GTSL2A2SDC.

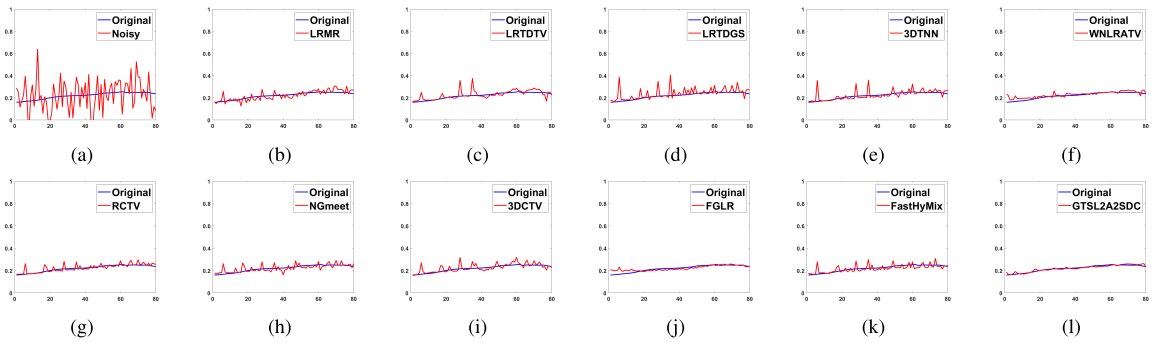


Fig. 8. Comparisons of the spectrum curves before and after denoising on the simulated PaviaC dataset under Case 4 at spatial position (170, 130). (a) Noisy image. (b) LRMR. (c) LRTDTV. (d) LRTDGS. (e) 3DTNN. (f) WNLRATV. (g) RCTV. (h) NGmeet. (i) 3DCTV. (j) FGLR. (k) FastHyMix. (l) GTSL2A2SDC.

remove the mixed noise and stripe noise. Particularly, LRTDGS and 3DTNN fail to remove the serious stripe noise. LRTDTV can better remove the mixed noise while smoothing the image details on the simulated Houston dataset, however,

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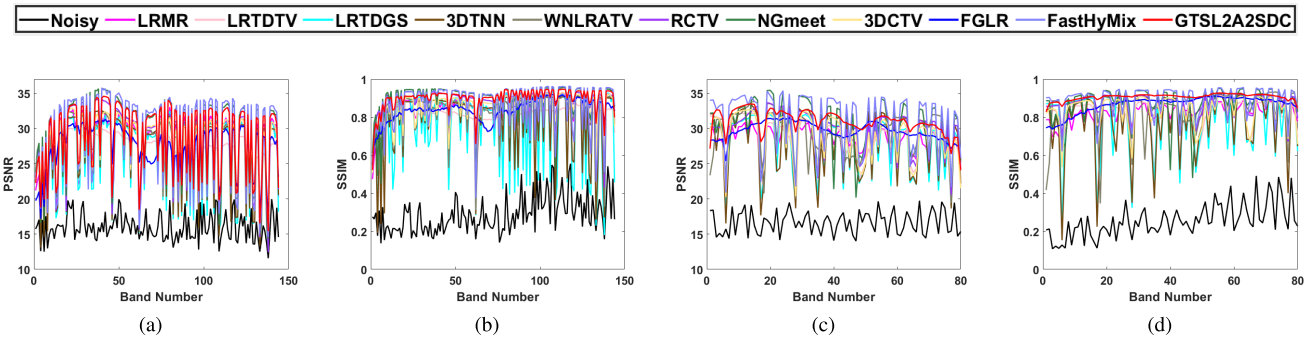


Fig. 9. Band-wise PSNR and band-wise SSIM of different approaches using simulated Houston and PaviaC datasets under Case 4. (a) Band-wise PSNR on Houston. (b) Band-wise SSIM on Houston. (c) Band-wise PSNR on PaviaC. (d) Band-wise SSIM on PaviaC.

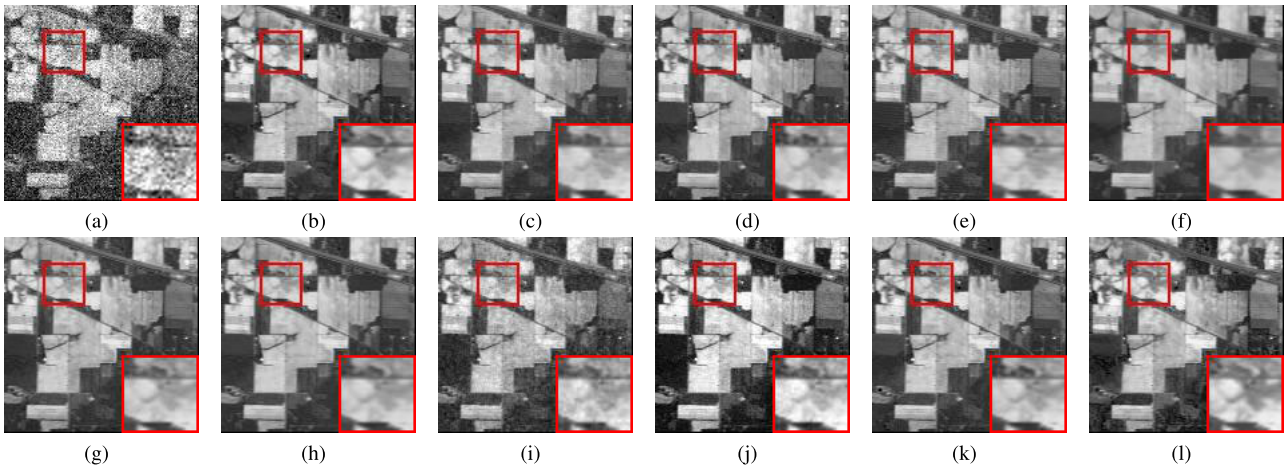


Fig. 10. Real denoising results of different approaches on the Indian pines dataset (band: 204). (a) Real noisy image. (b) LRMR. (c) LRTDTV. (d) LRTDGS. (e) 3DTNN. (f) WNLRATV. (g) RCTV. (h) NGmeet. (i) 3DCTV. (j) FGLR. (k) FastHyMix. (l) GTSL2A2SDC.

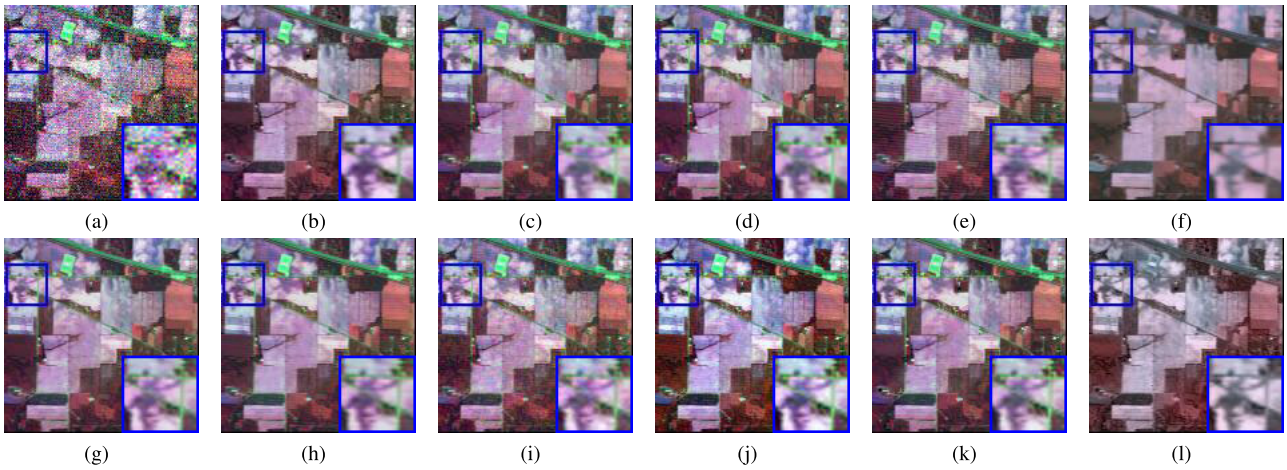


Fig. 11. Pseudo-color results of different denoising approaches on the real Indian pines dataset (bands: [103, 2, 204]). (a) Real noisy image. (b) LRMR. (c) LRTDTV. (d) LRTDGS. (e) 3DTNN. (f) WNLRATV. (g) RCTV. (h) NGmeet. (i) 3DCTV. (j) FGLR. (k) FastHyMix. (l) GTSL2A2SDC.

LRTDTV also fails to remove the serious stripe noise on the simulated PaviaC dataset. WNLRATV well removes most of the mixed noise and also smooths some details. RCTV better removes the mixed noise while preserving sharp details. FGLR better removes the stripe noise while still remaining some other types of mixed noise. NGmeet and FastHyMix both fail to remove the serious stripe noise. Meanwhile, LRMR, LRTDTV, LRTDGS, 3DTNN, WNLRATV, RCTV, NGmeet, 3DCTV, FGLR, and FastHyMix clearly cause the color distortions at different degrees. Apparently, from the aspects of removing the serious mixed noise particularly for the stripe noise, preserving image details and avoiding the color distortions, the proposed GTSL2A2SDC method shows the best visual denoising results.

Moreover, with the comparisons of spectrum curves before and after denoising on the simulated Houston and PaviaC

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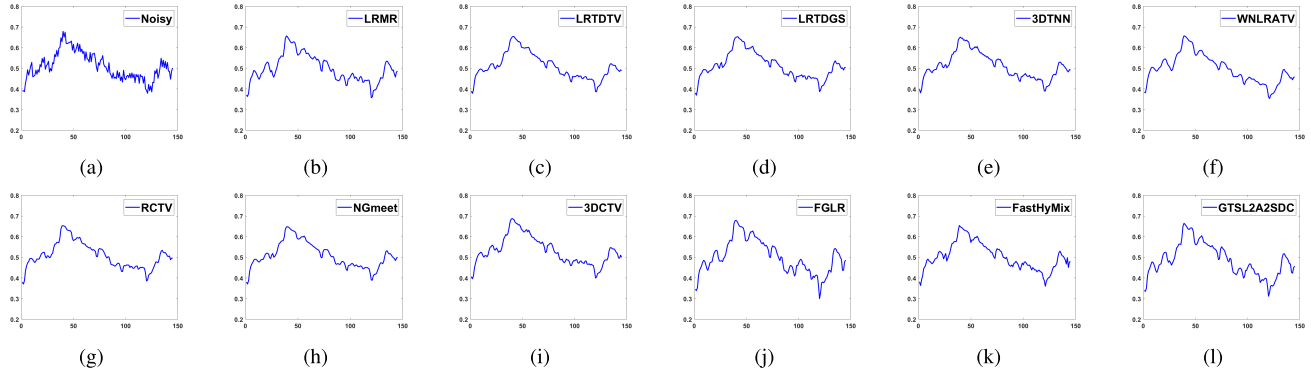


Fig. 12. Column mean curve results of different denoising approaches on the real Indian pines dataset (band: 109). (a) Real noisy image. (b) LRMR. (c) LRTDTV. (d) LRTDGS. (e) 3DTNN. (f) WNLRATV. (g) RCTV. (h) NGmeet. (i) 3DCTV. (j) FGLR. (k) FastHyMix. (l) GTSL2A2SDC.

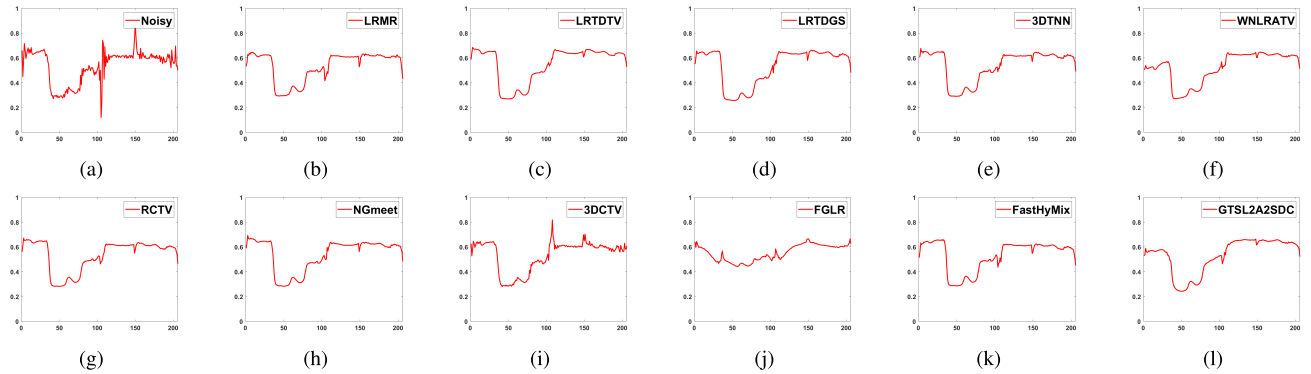


Fig. 13. Comparisons of the spectrum curves before and after denoising on the real Indian pines dataset at spatial position (65, 77). (a) Real noisy image. (b) LRMR. (c) LRTDTV. (d) LRTDGS. (e) 3DTNN. (f) WNLRATV. (g) RCTV. (h) NGmeet. (i) 3DCTV. (j) FGLR. (k) FastHyMix. (l) GTSL2A2SDC.

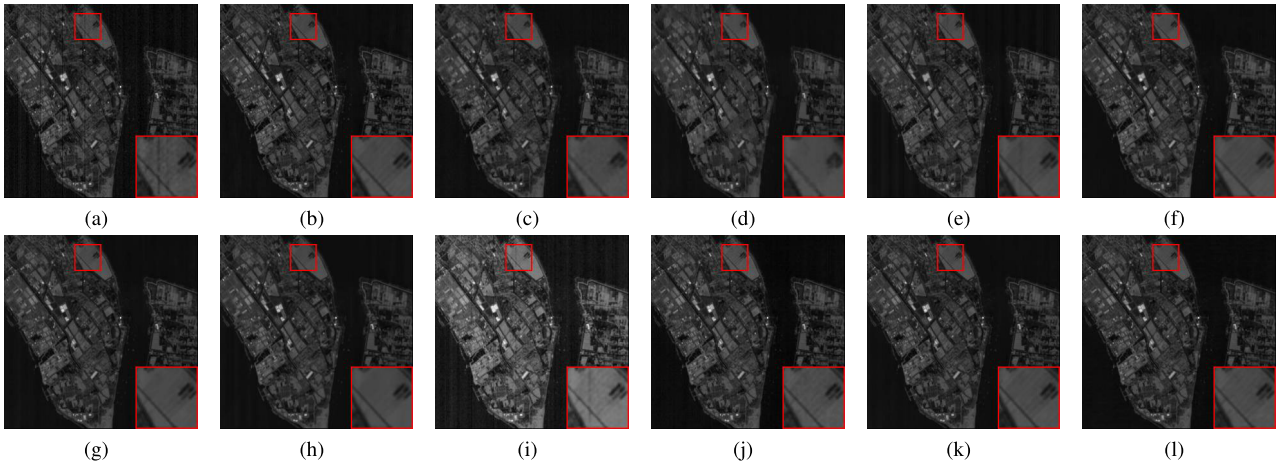


Fig. 14. Real denoising results of different approaches on the GaoFen-5 Shanghai dataset (band: 151). (a) Real noisy image. (b) LRMR. (c) LRTDTV. (d) LRTDGS. (e) 3DTNN. (f) WNLRATV. (g) RCTV. (h) NGmeet. (i) 3DCTV. (j) FGLR. (k) FastHyMix. (l) GTSL2A2SDC.

datasets shown in Figs. 5 and 8, we can clearly obtain that the proposed GTSL2A2SDC method shows the best spectrum preserving capacity.

Besides, we also show the PSNR and SSIM values of each HSI band (i.e., band-wise PSNR and band-wise SSIM) of different approaches using the simulated Houston and PaviaC datasets under Case 4 in Fig. 9, we can also clearly obtain that the proposed GTSL2A2SDC method generally performs more robust and stable band-wise SSIM and band-wise PSNR values than the remaining denoising approaches.

C. Real Denoising Experiments

In the real denoising experiments, we use the real Indian pines dataset (size: $145 \times 145 \times 206$) and the real Gaofen-5 Shanghai dataset (size: $300 \times 300 \times 155$) for directly denoising, where they both contain the serious mixed noise of different degrees, such as, Gaussian noise, impulse noise, deadlines, and stripe noise.

For visually validating the mixed noise removal performance of different approaches, we also show the detailed experimental results from the aspects of denoising results,

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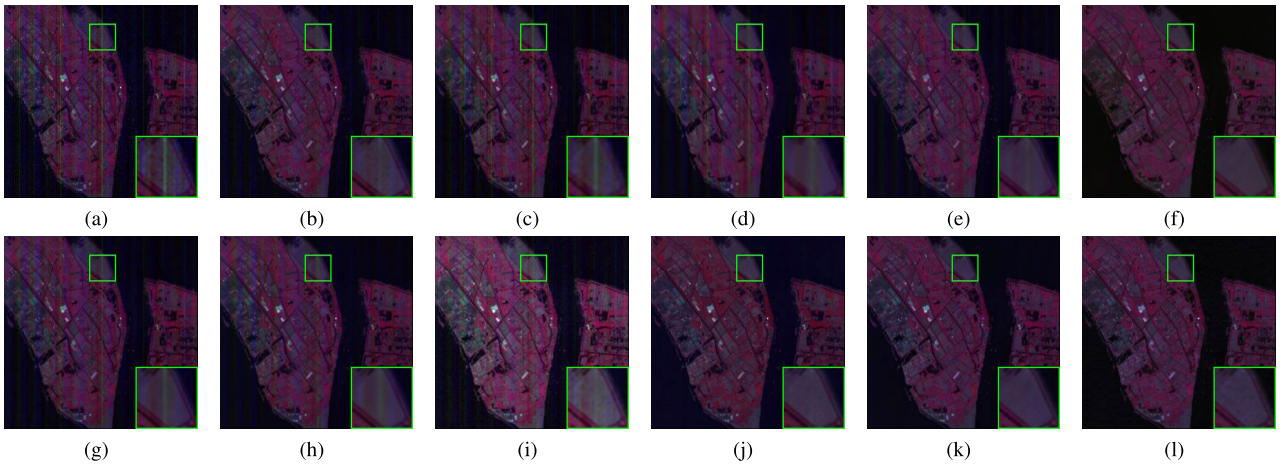


Fig. 15. Pseudo-color results of different denoising approaches on the real GaoFen-5 Shanghai dataset (bands: [13, 154, 155]). (a) Real noisy image. (b) LRMR. (c) LRTDTV. (d) LRTDGS. (e) 3DTNN. (f) WNLRATV. (g) RCTV. (h) NGmeet. (i) 3DCTV. (j) FGLR. (k) FastHyMix. (l) GTSL2A2SDC.

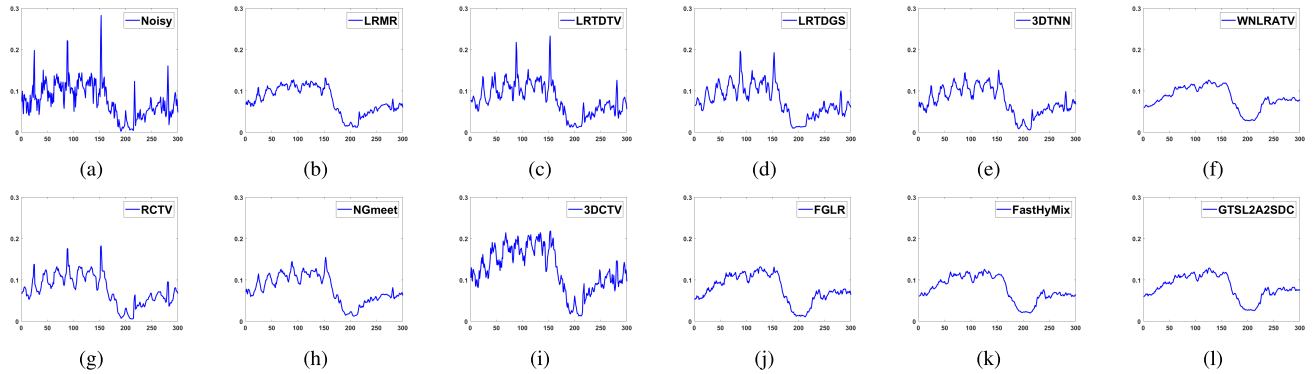


Fig. 16. Column mean curve results of different denoising approaches on the real GaoFen-5 Shanghai dataset (band: 154). (a) Real noisy image. (b) LRMR. (c) LRTDTV. (d) LRTDGS. (e) 3DTNN. (f) WNLRATV. (g) RCTV. (h) NGmeet. (i) 3DCTV. (j) FGLR. (k) FastHyMix. (l) GTSL2A2SDC.

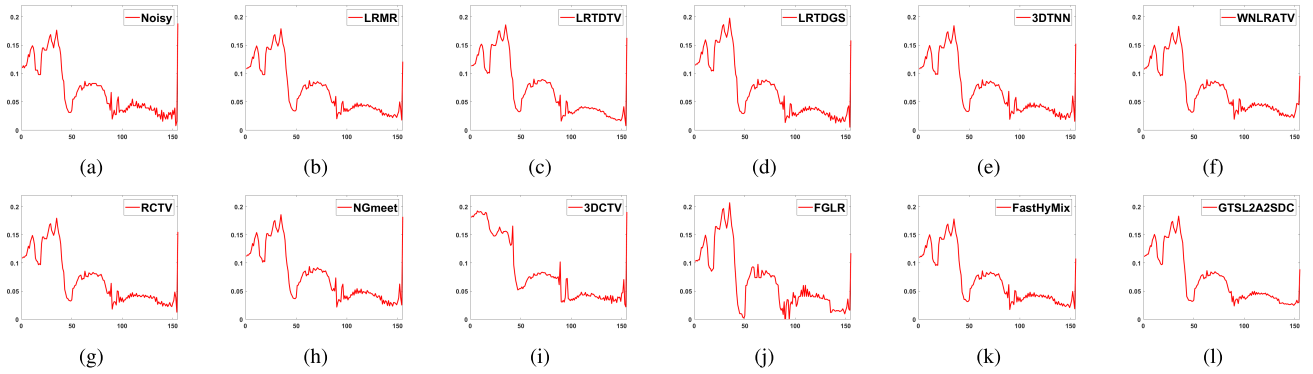


Fig. 17. Comparisons of the spectrum curves before and after denoising on the real GaoFen-5 Shanghai dataset at spatial position (121,189). (a) Real noisy image. (b) LRMR. (c) LRTDTV. (d) LRTDGS. (e) 3DTNN. (f) WNLRATV. (g) RCTV. (h) NGmeet. (i) 3DCTV. (j) FGLR. (k) FastHyMix. (l) GTSL2A2SDC.

pseudo-color results, column mean curves, and spectrum curves before and after denoising on real Indian pines and GaoFen-5 Shanghai datasets in Figs. 10–17, respectively.

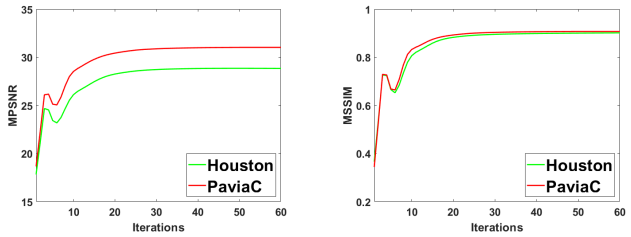
With the denoised images and zoomed details shown in Figs. 10, 11, 14, and 15, LRMR, LRTDTV, LRTDGS, 3DTNN, WNLRATV, RCTV, NGmeet, 3DCTV, FGLR, and FastHyMix visually either could not absolutely reduce the mixed noise or could not well preserve image details, and they also cause some color distortions at different degrees. Apparently, from the aspects of removing the serious mixed noise particularly for the stripe noise, preserving image details and avoiding the color distortions, the proposed GTSL2A2SDC method also shows the best visual denoising results.

Moreover, with the column mean curves shown in Figs. 12 and 16 where the selected bands both contain seriously mixed noise so that the column mean curves of real noisy images oscillate fast and largely, the proposed GTSL2A2SDC method can best smooth the column mean curves while reducing the oscillations. Meanwhile, with the comparisons of spectrum curves before and after denoising shown in Figs. 13 and 17

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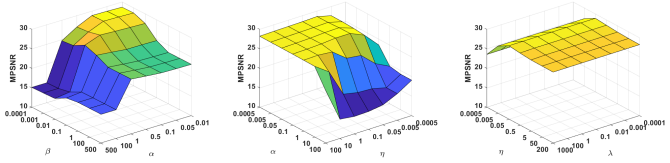
TABLE II
RUNNING TIME COMPARISON (IN SECONDS) OF COMPARED HSI DENOISING APPROACHES

Dataset	Size	LRMR	LRTDTV	LRTDGS	3DTNN	WNLRATV	RCTV	NGmeet	3DCTV	FGLR	FastHyMix	GTSL2A2SDC
Indian Pines	$145 \times 145 \times 206$	28.9	86.7	58.3	85.8	255.3	6.6	60.7	202.4	6.1	4.2	81.3
GaoFen-5 Shanghai	$300 \times 300 \times 155$	54.2	198.2	141.8	244.9	885.9	18.4	266.4	519.2	14.1	36.6	192.5



(a) (b)

Fig. 18. Convergence curves of MPSNR and MSSIM of GTSL2A2SDC using the simulated Houston and PaviaC datasets under Case 4. (a) MPSNR w.r.t iterations. (b) MSSIM w.r.t iterations.



(a) (b) (c)

Fig. 19. Discussion of parameters (β, α, η, and λ) of GTSL2A2SDC using the simulated PaviaC dataset under Case 4. (a) MPSNR w.r.t (β and α). (b) MPSNR w.r.t (α and η). (c) MPSNR w.r.t (η and λ).

where the spectrum curves of real noisy images in Figs. 13(a) and 17(a) also oscillate fast owing to the seriously mixed noise, the proposed GTSL2A2SDC method shows the best spectrum preserving capacity, which can best reduce the oscillations while better keeping minor changes of spectrum curves.

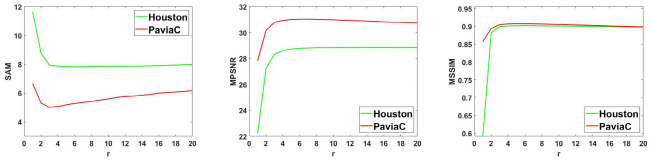
D. Convergence Analysis

To illustrate the convergence of the proposed GTSL2A2SDC algorithm, we particularly show the evolution curves of MPSNR and MSSIM with regard to (w.r.t) iterations of GTSL2A2SDC using the simulated Houston and PaviaC datasets under Case 4 in Fig. 18. As clearly displayed in Fig. 18, the MPSNR and MSSIM of GTSL2A2SDC become stable gradually and converge finally, which fully validates the convergence of GTSL2A2SDC.

E. Parameter Analysis

Then, the parameters β, α, η, and λ of the proposed GTSL2A2SDC model (16) are further analyzed.

Thus, Fig. 19 representatively shows the sensitivity of MPSNR w.r.t the parameters β, α, η, and λ of GTSL2A2SDC on the simulated PaviaC dataset under Case 4. As clearly shown in Fig. 19, the MPSNR results of GTSL2A2SDC are affected by different settings of β, α, η, and λ. In our experiments, we will choose $\beta \in [0.01, 1]$, $\alpha \in [0.01, 0.1]$, $\eta \in [0.005, 0.05]$, and $\lambda \in [0.01, 1]$ for the proposed GTSL2A2SDC method to show the best denoising performance.



(a) (b) (c)

Fig. 20. Discussion of r of GTSL2A2SDC using the simulated Houston and PaviaC datasets under Case 4. (a) SAM w.r.t r. (b) MPSNR w.r.t r. (c) MSSIM w.r.t r.

F. Gradient Subspace Dimension Analysis

Moreover, the effect of gradient subspace dimension r of GTSL2A2SDC is also analyzed, where $\text{rank}(\nabla_1 \mathbf{X}_{(3)}) = \text{rank}(\nabla_2 \mathbf{X}_{(3)}) = \text{rank}(\nabla_3 \mathbf{X}_{(3)}) = r$ (see Section II-D).

Thus, Fig. 20 representatively shows the SAM, MPSNR, and MSSIM results w.r.t r of GTSL2A2SDC using the simulated Houston and PaviaC datasets under Case 4. As clearly shown in Fig. 20, the SAM, MPSNR, and MSSIM results of GTSL2A2SDC are affected by different settings of r. Thus, in the experiments, we set $r = 7$ for the simulated Houston dataset and $r = 4$ for the simulated PaviaC dataset, respectively.

G. Running Time Analysis

Besides, Table II also gives the detailed results of running time of different denoising approaches on the real Indian pines and GaoFen-5 Shanghai datasets, where the running time result of FastHyMix is the shortest on the real Indian pines dataset and the running time result of FGLR is the shortest on the real GaoFen-5 Shanghai dataset. Although the running time results of GTSL2A2SDC are not the shortest, they are also acceptable owing to GTSL2A2SDC can perform the best mixed noise removal results.

H. Computational Complexity Analysis

Finally, the computational complexity of GTSL2A2SDC in Algorithm 1 is in detail analyzed as follows.

1) The complexity of $\mathcal{S}^{(k+1)}$ via (20) costs $O(n_1 n_2 n_3)$ per iteration.
2) The complexity of $\mathcal{B}^{(k+1)}$ via (23) costs $O(n_1 n_2 n_3 (\log(n_1 n_2 n_3)))$ per iteration.
3) The complexity of $\mathcal{G}^{(k+1)}$ via (25) costs $O(n_1 n_2 n_3 (\log(n_1 n_2 n_3) + \min(n_1, n_2)))$ per iteration.
4) The complexity of $\mathcal{P}_i^{(k+1)}$ via (28) costs $O(n_1 n_2 n_3)$ per iteration.
5) The complexity of $\mathcal{U}_i^{(k+1)}$ via (31) costs $O(n_1 n_2 n_3^2)$ per iteration.
6) The complexity of $\mathbf{V}_i^{(k+1)}$ via (34) costs $O(n_3 r^2)$ per iteration.

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7) The complexity of $\mathcal{X}^{(k+1)}$ via (37) costs $O(n_1n_2n_3(\log(n_1n_2n_3)))$ per iteration.
8) The complexity of $\mathcal{H}^{(k+1)}$ and $\mathcal{Q}^{(k+1)}$ via (38) cost $O(n_1n_2n_3)$ per iteration.

Thus, the computational complexity of GTSL2A2SDC in Algorithm 1 costs $O(n_1n_2n_3(\log(n_1n_2n_3) + \min(n_1, n_2)) + n_1n_2n_3(\log(n_1n_2n_3)) + n_1n_2n_3^2 + n_1n_2n_3 + n_3r^2)$ per iteration.

V. Conclusion

In this article, we have proposed a new GTSL2A2SDC method for HSI denoising and destriping method under the unified framework of tensor representation modeling. On one hand, based on the inherently directional and structural attributes of the stripe noise, we mainly investigate the mode-1 gradient tensor of stripe along the stripe direction which holds the desirable "zero plane" constraint as well as the mode-2 gradient tensor of stripe across the stripe direction which holds the desirable tensor low-fibered-rank attribute along the mode-3 spectral dimension. Therefore, we novelly propose the unified along-across stripe directional gradient tensor constraints for the stripe noise, which can simultaneously characterize the directional and structural attributes of the stripe noise. On the other hand, based on the desirably spectral low-rankness attributes of the multimode gradient tensors of HSI, we utilize the spectral low-rank factorization of the multimode gradient tensors of HSI to get the corresponding representation tensors, and particularly investigate the NSS-based low-rankness of the representation tensors under the multimode gradient tensor subspace low-rank learning framework. Moreover, we optimize the proposed GTSL2A2SDC model via the ADMM algorithm. At last, extensive experiments comprehensively validate the denoising and destriping capacity and superiority of GTSL2A2SDC.

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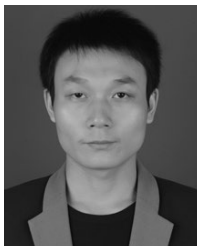
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